



GUJJA RAJU 2403A52018

```
import sys
!{sys.executable} -m pip install transformers datasets torch pandas scikit-learn
print("Libraries installed successfully.")
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages
```

Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages
Libraries installed successfully.

```
from datasets import load_dataset
```

```
# Load the 'imdb' dataset
```

```
imdb_dataset = load_dataset('imdb')
```

```
# Print the dataset to inspect its structure
```

```
print(imdb_dataset)
```

```
print("Dataset loaded successfully.")
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning: The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab. You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to access public datasets.
warnings.warn(

Warning: You are sending unauthenticated requests to the HF Hub. Please set a token as an environment variable or as a Colab secret.

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a token as an environment variable or as a Colab secret.

README.md: 0.00B [00:00, ?B/s]

plain_text/train-00000-of-00001.parquet: 0%| | 0.00/21.0M [00:00<?, ?B/s]

plain_text/test-00000-of-00001.parquet: 0%| | 0.00/20.5M [00:00<?, ?B/s]

plain_text/unsupervised-00000-of-00001.p(...): 0%| | 0.00/42.0M [00:00<?, ? examples/s]

Generating train split: 0%| | 0/25000 [00:00<?, ? examples/s]

Generating test split: 0%| | 0/25000 [00:00<?, ? examples/s]

Generating unsupervised split: 0%| | 0/50000 [00:00<?, ? examples/s]

```
DatasetDict({
  train: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  test: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  unsupervised: Dataset({
    features: ['text', 'label'],
    num_rows: 50000
  })
})
```

Dataset loaded successfully.

```
from transformers import DistilBertTokenizerFast
```

```
# 1. Load the DistilBertTokenizerFast tokenizer
```

```
tokenizer = DistilBertTokenizerFast.from_pretrained('distilbert-base-uncased')
```

```
# 2. Define a tokenization function
```

```
def tokenize_function(examples):
```

```
    return tokenizer(examples['text'], truncation=True, padding=True)
```

```
# 3. Apply this tokenization function to all splits of the imdb_dataset
```

```
tokenized_datasets = imdb_dataset.map(tokenize_function, batched=True, remove_
```

```
print("Dataset tokenization complete.")
```

```
print(tokenized_datasets)
```

tokenizer_config.json: 0%| | 0.00/48.0 [00:00<?, ?B/s]

```
vocab.txt: 0.00B [00:00, ?B/s]
```

```
tokenizer.json: 0.00B [00:00, ?B/s]
```

```
Map: 0%|          | 0/25000 [00:00<?, ? examples/s]
```

```
Map: 0%|          | 0/25000 [00:00<?, ? examples/s]
```

```
Map: 0%|          | 0/50000 [00:00<?, ? examples/s]
```

```
Dataset tokenization complete.
```

```
DatasetDict({
  train: Dataset({
    features: ['label', 'input_ids', 'attention_mask'],
    num_rows: 25000
  })
  test: Dataset({
    features: ['label', 'input_ids', 'attention_mask'],
    num_rows: 25000
  })
  unsupervised: Dataset({
    features: ['label', 'input_ids', 'attention_mask'],
    num_rows: 50000
  })
})
```

```
import torch
```

```
# 1. Extract the 'train' split
```

```
train_dataset = tokenized_datasets['train']
```

```
# 2. Extract the 'test' split (which is usually referred to as evaluation set)
```

```
eval_dataset = tokenized_datasets['test']
```

```
# 3. Print the structure of both datasets
```

```
print("Train Dataset Structure:", train_dataset)
```

```
print("\nEvaluation Dataset Structure:", eval_dataset)
```

```
print("Datasets prepared successfully.")
```

```
Train Dataset Structure: Dataset({
  features: ['label', 'input_ids', 'attention_mask'],
  num_rows: 25000
})
```

```
Evaluation Dataset Structure: Dataset({
  features: ['label', 'input_ids', 'attention_mask'],
  num_rows: 25000
})
```

Datasets prepared successfully.

```
from transformers import DataCollatorWithPadding
from torch.utils.data import DataLoader
```

```
# 1. Set the format of both train_dataset and eval_dataset to 'torch'
train_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask'])
eval_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask'])
```

```
# 2. Instantiate DataCollatorWithPadding
data_collator = DataCollatorWithPadding(tokenizer=tokenizer)
```

```
# 3. Create DataLoader instances
batch_size = 8 # Adjust based on GPU memory
train_dataloader = DataLoader(train_dataset, shuffle=True, batch_size=batch_size)
eval_dataloader = DataLoader(eval_dataset, batch_size=batch_size, collate_fn=data_collator)
```

```
print("PyTorch Datasets and DataLoaders created successfully.")
print(f"Train DataLoader batch size: {train_dataloader.batch_size}")
print(f"Evaluation DataLoader batch size: {eval_dataloader.batch_size}")
```

PyTorch Datasets and DataLoaders created successfully.
Train DataLoader batch size: 8
Evaluation DataLoader batch size: 8

```
from transformers import DistilBertForSequenceClassification
```

```
# 1. Load the pre-trained DistilBertForSequenceClassification model
# 2. Specify num_labels=2 for binary sentiment classification (positive/negative)
model = DistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased')
```

```
print("Pre-trained DistilBertForSequenceClassification model loaded successfully.")
# 3. Optionally, print the model architecture to verify
print(model)
```

```
config.json: 0%|          | 0.00/483 [00:00<?, ?B/s]
```

model.safetensors: 0%| | 0.00/268M [00:00<?, ?B/s]

Loading weights: 0%| | 0/100 [00:00<?, ?it/s]

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status
-----+-----	-----+-----
vocab_layer_norm.weight	UNEXPECTED
vocab_projector.bias	UNEXPECTED
vocab_layer_norm.bias	UNEXPECTED
vocab_transform.bias	UNEXPECTED
vocab_transform.weight	UNEXPECTED
pre_classifier.weight	MISSING
pre_classifier.bias	MISSING
classifier.weight	MISSING
classifier.bias	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture
- MISSING :those params were newly initialized because missing from the che

Pre-trained DistilBertForSequenceClassification model loaded successfully.

```
DistilBertForSequenceClassification(
  (distilbert): DistilBertModel(
    (embeddings): Embeddings(
      (word_embeddings): Embedding(30522, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (transformer): Transformer(
      (layer): ModuleList(
        (0-5): 6 x TransformerBlock(
          (attention): DistilBertSelfAttention(
            (q_lin): Linear(in_features=768, out_features=768, bias=True)
            (k_lin): Linear(in_features=768, out_features=768, bias=True)
            (v_lin): Linear(in_features=768, out_features=768, bias=True)
            (out_lin): Linear(in_features=768, out_features=768, bias=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (sa_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
          (ffn): FFN(
            (dropout): Dropout(p=0.1, inplace=False)
```

```

        (lin1): Linear(in_features=768, out_features=3072, bias=True)
        (lin2): Linear(in_features=3072, out_features=768, bias=True)
        (activation): GELUActivation()
    )
    (output_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
)
)
)
)
(pre_classifier): Linear(in_features=768, out_features=768, bias=True)
(classifier): Linear(in_features=768, out_features=2, bias=True)
(dropout): Dropout(p=0.2, inplace=False)
)

```

```
from transformers import DistilBertForSequenceClassification
```

```

# 1. Load the pre-trained DistilBertForSequenceClassification model
# 2. Specify num_labels=2 for binary sentiment classification (positive/negative)
model = DistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased')

```

```

print("Pre-trained DistilBertForSequenceClassification model loaded successfully")
# 3. Optionally, print the model architecture to verify
print(model)

```

```
Loading weights: 0%|          | 0/100 [00:00<?, ?it/s]
```

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
```

Key	Status
-----+-----	-----+-----
vocab_layer_norm.weight	UNEXPECTED
vocab_projector.bias	UNEXPECTED
vocab_layer_norm.bias	UNEXPECTED
vocab_transform.bias	UNEXPECTED
vocab_transform.weight	UNEXPECTED
pre_classifier.weight	MISSING
pre_classifier.bias	MISSING
classifier.weight	MISSING
classifier.bias	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture
- MISSING :those params were newly initialized because missing from the checkpoint

```
Pre-trained DistilBertForSequenceClassification model loaded successfully.
```

```

DistilBertForSequenceClassification(
  (distilbert): DistilBertModel(
    (embeddings): Embeddings(

```

```

(word_embeddings): Embedding(30522, 768, padding_idx=0)
(position_embeddings): Embedding(512, 768)
(LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
(dropout): Dropout(p=0.1, inplace=False)
)
(transformer): Transformer(
  (layer): ModuleList(
    (0-5): 6 x TransformerBlock(
      (attention): DistilBertSelfAttention(
        (q_lin): Linear(in_features=768, out_features=768, bias=True)
        (k_lin): Linear(in_features=768, out_features=768, bias=True)
        (v_lin): Linear(in_features=768, out_features=768, bias=True)
        (out_lin): Linear(in_features=768, out_features=768, bias=True)
        (dropout): Dropout(p=0.1, inplace=False)
      )
      (sa_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
      (ffn): FFN(
        (dropout): Dropout(p=0.1, inplace=False)
        (lin1): Linear(in_features=768, out_features=3072, bias=True)
        (lin2): Linear(in_features=3072, out_features=768, bias=True)
        (activation): GELUActivation()
      )
      (output_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
    )
  )
)
)
(pre_classifier): Linear(in_features=768, out_features=768, bias=True)
(classifier): Linear(in_features=768, out_features=2, bias=True)
(dropout): Dropout(p=0.2, inplace=False)
)

```

```

from transformers import TrainingArguments

```

```

# 2. Instantiate TrainingArguments

```

```

training_args = TrainingArguments(
    output_dir='./results',
    num_train_epochs=3,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    warmup_steps=500,
    weight_decay=0.01,
    logging_dir='./logs',
    logging_steps=10,
    # Temporarily removing evaluation_strategy and save_strategy due to TypeError
    # and disabling load_best_model_at_end to resolve the subsequent ValueError
    # Ideally, these would be set to 'epoch' and match.
    load_best_model_at_end=False, # Changed to False to bypass ValueError
    metric_for_best_model='accuracy',

```



```

        report_to='none'
    )

print("Training Arguments defined successfully.")
# 3. Print the instantiated TrainingArguments object to verify the settings
print(training_args)

`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD`

```

```

Training Arguments defined successfully.
TrainingArguments(
  accelerator_config={'split_batches': False, 'dispatch_batches': None, 'even_batches': True},
  adam_beta1=0.9,
  adam_beta2=0.999,
  adam_epsilon=1e-08,
  auto_find_batch_size=False,
  average_tokens_across_devices=True,
  batch_eval_metrics=False,
  bf16=False,
  bf16_full_eval=False,
  data_seed=None,
  dataloader_drop_last=False,
  dataloader_num_workers=0,
  dataloader_persistent_workers=False,
  dataloader_pin_memory=True,
  dataloader_prefetch_factor=None,
  ddp_backend=None,
  ddp_broadcast_buffers=None,
  ddp_bucket_cap_mb=None,
  ddp_find_unused_parameters=None,
  ddp_timeout=1800,
  debug=[],
  deepspeed=None,
  disable_tqdm=False,
  do_eval=False,
  do_predict=False,
  do_train=False,
  enable_jit_checkpoint=False,
  eval_accumulation_steps=None,
  eval_delay=0,
  eval_do_concat_batches=True,
  eval_on_start=False,
  eval_steps=None,
  eval_strategy=IntervalStrategy.NO,
  eval_use_gather_object=False,
  fp16=False,
  fp16_full_eval=False,
  fsdp=[],

```

```
fsdp_config={'min_num_params': 0, 'xla': False, 'xla_fsdp_v2': False, 'xla_fs
full_determinism=False,
gradient_accumulation_steps=1,
gradient_checkpointing=False,
gradient_checkpointing_kwargs=None,
greater_is_better=True,
group_by_length=False,
hub_always_push=False,
hub_model_id=None,
hub_private_repo=None,
hub_revision=None,
hub_strategy=HubStrategy.EVERY_SAVE,
hub_token=<HUB_TOKEN>,
ignore_data_skip=False,
include_for_metrics=[],
include_num_input_tokens_seen=no,
label_names=None,
label_smoothing_factor=0.0,
learning_rate=5e-05,
length_column_name=length,
liger_kernel_config=None,
load_best_model_at_end=False,
local_rank=-1,
log_level=passive,
log_level_replica=warning,
log_on_each_node=True,
logging_dir=./logs,
logging_first_step=False,
logging_nan_inf_filter=True,
logging_steps=10,
logging_strategy=IntervalStrategy.STEPS,
lr_scheduler_kwargs=None,
lr_scheduler_type=SchedulerType.LINEAR,
max_grad_norm=1.0,
max_steps=-1,
metric_for_best_model=accuracy,
neftune_noise_alpha=None,
num_train_epochs=3,
optim=OptimizerNames.ADAMW_TORCH_FUSED,
optim_args=None,
optim_target_modules=None,
output_dir=./results,
parallelism_config=None,
per_device_eval_batch_size=8,
per_device_train_batch_size=8,
prediction_loss_only=False,
project=huggingface,
push_to_hub=False,
```

```

remove_unused_columns=True,
report_to=[],
restore_callback_states_from_checkpoint=False,
resume_from_checkpoint=None,
run_name=None,
save_on_each_node=False,
save_only_model=False,
save_steps=500,
save_strategy=SaveStrategy.STEPS,
save_total_limit=None,
seed=42,
skip_memory_metrics=True,
tf32=None,
torch_compile=False,
torch_compile_backend=None,
torch_compile_mode=None,
torch_empty_cache_steps=None,
trackio_space_id=trackio,
use_cache=False,
use_cpu=False,
use_liger_kernel=False,
warmup_ratio=None,
warmup_steps=500,
weight_decay=0.01,
)

```

```

import sys
!{sys.executable} -m pip install evaluate
print(" 'evaluate' library installed successfully.")

```

Collecting evaluate

```

  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: datasets>=2.0.0 in /usr/local/lib/python3.12/dist-packages (2.18.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (1.26.4)
Requirement already satisfied: dill in /usr/local/lib/python3.12/dist-packages (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: requests>=2.19.0 in /usr/local/lib/python3.12/dist-packages (2.32.3)
Requirement already satisfied: tqdm>=4.62.1 in /usr/local/lib/python3.12/dist-packages (4.67.1)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (3.5.0)
Requirement already satisfied: multiprocessing in /usr/local/lib/python3.12/dist-packages (3.0.4)
Requirement already satisfied: fsspec>=2021.05.0 in /usr/local/lib/python3.12/dist-packages (2025.3.0)
Requirement already satisfied: huggingface-hub>=0.7.0 in /usr/local/lib/python3.12/dist-packages (0.27.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (24.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (3.16.1)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (17.0.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (6.0.2)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (3.11.11)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (0.27.2)

```

```

Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages
Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)

```

```

[2K  [90m-----[0m [32m84.1/84.1 kB

```

```

[?25hInstalling collected packages: evaluate

```

```

Successfully installed evaluate-0.4.6

```

```

'evaluate' library installed successfully.

```

```

import evaluate

```

```

import numpy as np

```

```

# Define a function to compute evaluation metrics

```

```

def compute_metrics(eval_pred):

```

```

    predictions, labels = eval_pred

```

```

    # Apply np.argmax to predictions to get the predicted class labels.

```

```

    predictions = np.argmax(predictions, axis=1)

```

```

    # Load the metrics

```

```

    accuracy_metric = evaluate.load('accuracy')

```

```

    precision_metric = evaluate.load('precision')

```

```

    recall_metric = evaluate.load('recall')

```

```

    f1_metric = evaluate.load('f1')

```

```

    # Compute each metric

```

```

accuracy = accuracy_metric.compute(predictions=predictions, references=labels)
precision = precision_metric.compute(predictions=predictions, references=labels)
recall = recall_metric.compute(predictions=predictions, references=labels)
f1 = f1_metric.compute(predictions=predictions, references=labels, average='micro')

return {
    'accuracy': accuracy,
    'precision': precision,
    'recall': recall,
    'f1': f1
}

```

```
print("Compute metrics function defined successfully.")
```

Compute metrics function defined successfully.

```
from transformers import Trainer
```

```
# Initialize the Trainer
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=eval_dataset,
    compute_metrics=compute_metrics,
    data_collator=data_collator,
)
```

```
print("Trainer initialized successfully.")
```

Trainer initialized successfully.

```
print("Starting model training...")
```

```
trainer.train()
```

```
print("Model training complete.")
```

```
print("Starting model evaluation...")
```

```
results = trainer.evaluate()
```

```
print("Model evaluation complete.")
```

```
print(results)
```

Starting model training...

```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning:
  super().__init__(loader)

```

<div>

```
<progress value='286' max='9375' style='width:300px; height:20px; vertical-align:top'>
  [ 286/9375 1:59:08 < 63:33:10, 0.04 it/s, Epoch 0.09/3]
```

</div>

```
<table border="1" class="dataframe">
```

```
Step Training Loss 10 0.696825 20 0.696316 30 0.700587 40 0.696785 50
0.690993 60 0.691060 70 0.690644 80 0.682843 90 0.676282 100 0.690169 110
0.646369 120 0.654151 130 0.610866 140 0.501066 150 0.437934 160 0.370622
170 0.240504 180 0.372620 190 0.426576 200 0.522337 210 0.299034 220
0.271981 230 0.401184 240 0.307400 250 0.252992 260 0.487919 270 0.571100
280 0.421148
```